

# Note 1.1 Functions

## Functions; Domain and Range

- **Definition of function:**

A function  $f$  is a corresponding relationship that for any element  $x \in D$  there will be a unique element  $f(x) \in R$  to  $x$ .

- **Domain ( denoted by  $D$  or  $D(f)$  )**

- Natural Domain: the domain that makes functions meaningful.
- Notation: Set notation  $(\{x|x > 1\})$ , Interval notation  $((1, +\infty))$ .

- **Range ( denoted by  $R$  or  $R(f)$  )**

- Rules to judge whether two functions are the same:  
same domain and the same rule.

## The Vertical Line Test For a Function

- No vertical line can intersect the graph of a function more than once.

## Odd and Even Functions

- Geometrical Properties
- Arithmetic of these functions
- Property: If  $f(x) = 0$  and  $af(x) + bf\left(\frac{1}{x}\right) = \frac{c}{x}, x \neq 0; a, b, c$  are constants;  $|a| = |b|$ , then  $f(x)$  is an odd function.

## Monotonicity of a function

- Increasing and decreasing.

## Other Types of functions

- Piecewise-Defined Functions:
  - Absolute value function:  $f(x) = |x|$
  - Greatest Integer Function(integer floor function) ( $f(x) = \lfloor x \rfloor$ )
  - Least Integer Function(integer ceiling function) ( $f(x) = \lceil x \rceil$ )
- constant function:  $f(x) = c$
- Dirichlet Function:  $f(x) = \begin{cases} 1, & x \text{ is rational numbers.} \\ 0, & x \text{ is not rational.} \end{cases}$
- Linear Function:  $f(x) = kx + b$ , Proportional:  $y = kx$
- Power Function:  $f(x) = x^a$

- exponential and logarithmic functions: (  $f(x) = e^x$ ,  $f(x) = \ln x$  )
- polynomials
- rational function: quotient of two polynomials. If the degree of numerator is higher than denominator's, it is improper rational functions. Otherwise it is proper rational functions. Generally, we can split a improper rational function into the sum of a polynomial and a proper rational function.